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Value Sensitive Design Approach for Human-Centered Automated Chest X-Ray Report Generation Using Large Language Models and Computer Vision

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Declaration

I hereby certify that the material, which I now submit for assessment on the programmes of study leading to the award of Master of Science, is entirely my own work and has not been taken from the work of others except to the extent that such work has been cited and acknowledged within the text of my work. No portion of the work contained in this thesis has been submitted in support of an application for another degree or qualification to this or any other institution.



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January 13, 2026

Abstract

The growing clinical workload in radiology has drawn more attention to automated chest radiograph report generation systems. Deep learning and large language models (LLMs) are promising areas of research, but the issues of clinical reliability, transparency, and trustworthiness still significantly challenge these fields. For instance, quite a few existing systems produce grammatically correct reports while lacking the proper visual evidence grounding or human-centered and ethical design principles considerations. This research aims to help with these challenges by opening up a path to the integration of Value Sensitive Design (VSD) and explainability mechanisms into an AI-powered radiology reporting system. The main goal of this study is to create and test a human-centered automated chest X-ray report generation pipeline that does not compromise performance for transparency, clinician oversight, and ethical considerations. Two different vision-language models were built and compared: one that is based on ViT-GPT2 and the other, R2GenCMN, which is an extension of the domain and uses multi-view image encoding along with a memory-driven language decoder. The choice of model was based on quantitative evaluations using ROUGE-L metrics, diversity analysis, and qualitative clinical comparisons. Hyperparameter tuning and prolonged training were done to evaluate how optimization decisions affect report quality and generalization.

As a result, R2GenCMN is said to be the model that has consistently surpassed its baseline counterpart, resulting in reports that have a greater semantic alignment with the ground truth and more clinically structured outputs. The prolonged training improved disease-specific descriptions and reduced the production of ultra-general outputs, though some template repetition was found. To clarify the process, Grad-CAM-based visual explanations were used to spotlight the areas in the image that influenced the particular report generation. In addition, a physician-focused Streamlit interface was created to facilitate report visualization, explainability, and human-in-the-loop editing.

Thus, this research proves that the integration of domain-specific vision-language models with explainability and Human-centered design principles can greatly increase the usability and trustworthiness of automated radiology reporting systems. The facilitative role of these systems in clinical practice should be seen rather than that of the machines making autonomous decisions.

Keywords

Automated Chest X-Ray Report Generation , Value Sensitive Design, ViT–GPT2, R2GenCMN, XAI,

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The authors of the datasets, tools, and open-source software that are accessible publicly and were utilized during the research, especially the ones in the field of medical imaging and vision-language modeling, are to be recognized. Their commitment to open research practices and reproducibility made this study feasible.

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1 Introduction

1.1 Introduction

Artificial Intelligence (AI) has gradually developed its role in the medical sector, being a supporter of clinical decisions, e.g., the choice of treatment, and the like, improving operational efficiency and cutting down the workload of practitioners. During the procedure of medical imaging, chest X-rays are among the most wanted and used methods concerning diagnosis, mainly because of their low cost, speed, and good diagnostic value in identifying thoracic diseases like pneumonia, cardiomegaly, pleural effusion, and lung opacities. However, doctors still need considerable training and expertise to accurately refer to X-ray images' interpretation and reporting. The process is time-consuming, and radiologists' availability is often the determining factor for the entire X-ray workflow from taking and interpreting images to issuing reports.

One of the key applications of artificial intelligence in radiology is the automatic generation of reports, enabled by advances in computer vision and vision–language modeling, which allow clinical narratives to be generated directly from medical images. The recent development of so-called "multimodal deep learning models" is an increasing trend in the generation of reports, since they are able to produce such texts that are fluent and coherent. However, the clinicians' trust in these models has not been completely gained. Concerns about the models' clinical, reliability, and ethical alignment still exist. In the medical sector where risks are high, incorrect or misleading outputs can easily result in the loss of trust in the clinician and the endangerment of the patient's safety.

A human-centered approach to the automatic generation of chest X-ray report, which integrates Value Sensitive Design (VSD) principles into the development and evaluation of multimodal AI systems, is suggested in this thesis as a means to tackle the aforementioned difficulties. 7.

1.2 Background Information

In the beginning, automated radiology reporting systems utilized rule-based approaches and predefined text templates filled in with detected image features. These systems guaranteed grammatical correctness, yet they were inflexible and incapable of capturing subtle or unusual clinical findings. Among others, the deep learning era opened the door for more expressive representations, particularly the use of Convolutional Neural Networks (CNN) for visual feature extraction and transformer-based architectures for language modeling.

CheXNet, a CNN-based model, and others like it have already accomplished near-radiologist-level performance in the classification of chest X-ray images, although the generalization of these models was still limited to the prediction of labels instead of the generation of narrative reports. In order to overcome this restriction, encoder-decoder architectures were developed that integrated CNN image encoders with Recurrent Neural Networks or Transformer-based decoders to produce free-text radiology reports. More

recently, the development of vision-language models that share multimodal representations has positively impacted linguistic fluency and contextual coherence.

In spite of all these technical advancements, the majority of the existing methods still rely heavily on linguistic similarity metrics such as BLEU and ROUGE for evaluation. Previous research has shown that these metrics only weakly correlate with clinical correctness and factual accuracy in the medical reporting tasks. In addition, factors such as explainability, human oversight, and ethical governance are often given minimal consideration, hence the limitations of these systems for clinical acceptance.

1.3 Motivation

Despite the promise of decreased clinician workload and uniform reporting practice, the acceptance of automated report generation systems in healthcare is mainly impeded by lack of trust, transparency, accountability, and human intervention. The radiologists should have an understanding of the AI-driven outputs and should be able to criticize them especially when the outputs can affect clinical decisions that follow.

The regulatory viewpoint regards automated radiology reporting systems as high-risk AI systems under the proposed European Union Artificial Intelligence Act owing to their role in healthcare decision support. The Act prescribes to the AI systems requiring transparency, traceability, and meaningful human oversight. The Assessment List for Trustworthy Artificial Intelligence (ALTAI) also points out the principles of human agency, technical robustness, and accountability as necessary for AI to be regarded as trustworthy and thus, deployed.

This research aims to eliminate the gap separating the high-performance multimodal AI models and the human-centered design methodologies by incorporating the ethical and regulatory issues directly into the system design. The use of a Value Sensitive Design approach is to ensure that the automated radiology report generation systems are in the service of the clinician's trust, patient safety, and compliance with the regulatory requirements.

1.4 Scope of the Research and Relevance to Human-Centered AI

Automated chest X-ray report generation using multimodal AI models trained on publicly available and de-identified datasets is the main topic of this research. The entire process is an academic one that includes data and model preparation, evaluation, and explainability rather than clinical application or decision-making support.

This project is part of the larger field of Human-Centered Artificial Intelligence, which advocates for the compatibility of AI with human values, user requirements, and social standards. Transparency, human control, fairness, and accountability are among the core principles that are highlighted and through which it is guaranteed that the system to be developed will support clinical practice rather than supplanting professionals' judgment.

1.5 Research Questions and Objectives

This thesis is guided by the following research questions:

1. How can Value Sensitive Design be applied to guide the development of an AI-driven radiology report generation system?

2. How can Large Language Models be effectively fine-tuned for domain-specific chest X-ray report generation?

3. To what extent can explainability mechanisms and a clinician-centered interface enhance transparency, usability, and trust in AI-generated radiology reports?

The primary objective of this research is to design, implement, and evaluate a human-centered multimodal AI system for chest X-ray report generation that integrates ethical considerations into both technical development and system evaluation.

1.6 Structure of the Thesis

The organization of the rest of the thesis is as follows. A comprehensive overview of the literature concerning the topics of automated radiology reporting, multimodal AI models, and in particular, the ethics of AI in healthcare is provided in Chapter 2. Chapter 3 then goes on to explain the process of data collection, the characteristics of the dataset, the steps of preprocessing, and the exploratory analysis. Next, chapter 4 describes in detail the methodolgy, which includes both baseline and multimodal architectures. Explainable AI techniques Human centered aspects and the ethical relevance along with sustainability and regulatory alignment are discussed in chapter 5. Chapter 6 looks into the future work. Lastly, Chapter 7 provides a summary of the thesis along with future research directions.

2 Literature Review

2.1 Overview of Previous Works

The automatic radiology report writing has been a topic of great interest and research ever since the day when the presentation of the imaging data was made great by the use of Computer Vision (CV) and Natural Language Processing (NLP) techniques. Initially, the strategies were based on the use of a set of rules and templates for the reports where the features found in the images would be converted into text based on a predetermined structure. These systems, however, ensured that they were grammatically correct and the users were able to interpret them, but they were not flexible enough to represent and consequently not able to diagnose complex or atypical clinical situations.

The advent of deep learning techniques created a huge breakthrough in the area of medical image analysis. Convolutional Neural Networks (CNNs) like CheXNet and DenseNet-121 have reported almost the same performance as radiologists in the case of classification of chest X-rays where pneumonia and other lung-related problems were the main topics. However, these models only dealt with the classification of images and did not touch upon the important issue of narrative diagnostic reports, which is crucial for the clinical workflow.

Encoder-decoder architectures were proposed to fix this issue; they piled CNN-based image encoders on the top of language generation models. TieNet achieved by integrating visual features with textual embeddings hence jointly did the tasks of disease classification and report generation. Later on, models like R2Gen and memory-driven transformers brought about the fluency and coherence of the reports to a remarkable extent by including attention mechanisms and structured memory representations. Even with these advances, the majority of the systems used surface-level text similarity metrics like BLEU and ROUGE to optimize their performance.

In recent studies, the emphasis has been placed on the use of multimodal vision-language models that represent the same concepts in both the visual and text domains. The use of transformer-based architecture has led to better alignment between the image features and the linguistic tokens thus making it easier to generate reports that are more human-like and with higher expressiveness. The models trained on large-scale data, such as BLIP-2, LLaVA, and MedPaLM-2, have not only proven their large-scale applicability but have also been explored within the medical imaging domain for the tasks of captioning and report generation. These models have their capability in generating fluent and context-aware outputs owing to their extensive pretraining on different datasets of varied characteristics.

Technical developments have not only led to the continuous consideration of ethical and human-centered aspects in AI for healthcare, but also made it necessary to do so. The Value Sensitive Design (VSD) is a method that allows one to integrate human values into the design of a system by conceptual, empirical, and technical investigations. In the areas of patient applications and decision-support systems, VSD has been utilized,

reinforcing the need for transparency, trust, and accountability in these areas. The regulation measures, like the European Union Artificial Intelligence Act and the Assessment List for Trustworthy Artificial Intelligence (ALTAI), have also taken it to the next level by demanding maintenance of the three key concepts (transparency, human oversight, and accountability) in the case of high-risk AI systems.

2.2 Identified Gaps

In spite of the huge strides made, automated radiology report generation systems still have many drawbacks in their current form. To start with, the use of linguistic similarity metrics like BLEU and ROUGE is still widespread in the evaluation of many studies. These metrics analyze the amount of textual overlap between the produced and the original reports, but they show weak correlation with clinical correctness and factual accuracy in medical reporting tasks. Therefore, it is possible for the models to receive very high scores on the metrics while they still mislead the clinicians or produce hallucinated content.

Next, the large multimodal models are often very powerful but at the same time very difficult to understand and where their reasoning is. They do produce very high-quality texts, but the doctors are not informed at all of the models' reasoning and hence the reports cannot be trusted. The lack of an explanation can make it very difficult for the hospitals to adopt such technologies in their working environments, where safety is the priority.

Ethical and legal issues are mostly treated in a reactive manner and after the fact, rather than being incorporated into the design of the system in a preventive way. While it is very clear from the likes of VSD, ALTAI, and the EU AI Act that the principles of trustworthy AI exist, there are hardly any studies that work out the principles in technical or organizational design requirements for multimodal medical AI systems in very detailed and concrete manner. The human oversight mechanisms that are found are usually very vaguely defined and not subjected to systematic assessment.

Lastly, the current research very seldom takes the clinician's viewpoint during the model development. The lack of practitioners' involvement results in the inability of the current systems to solve the key issues of usability, trust, and integration with the workflow in the real world. These gaps highlight the need for a more holistic approach that combines technical performance with human-centered and ethical design considerations.

2.3 Research Novelty

This dissertation deals with the recognized gaps by utilizing a Value Sensitive Design approach to the generation of multimodal chest X-ray reports. In contrast to previous studies that mainly concentrate on technical performance, this research deliberately incorporates stakeholder values—like transparency, trust, fairness, accountability, and human oversight—into the system design and evaluation procedure.

The suggested method merges a modular multimodal architecture with explainability techniques and human-in-the-loop systems, which guarantees that doctors still have a substantial control over the AI-generated outputs. Ethical and regulatory principles are implemented through quantifiable indicators such as a Trust Index and a Value Compliance Index, which allow for the systematic evaluation of both technical effectiveness and ethical alignment.

Through the combination of multimodal AI modeling with Value Sensitive Design, regulatory awareness, and clinician-centered evaluation, this research delivers a novel,

human-centered framework for automated radiology report generation. The proposed methodology not only aspires towards the creation of AI systems that are technically sophisticated but also trustworthy, transparent, and in sync with clinical and societal values.

3 Data Collection and Preparation

3.1 Data Collection

The research reported here employs the Indiana University Chest X-Ray (IU X-Ray) dataset which was acquired through Kaggle. This dataset is a fully de-identified and openly accessible one that can be used for research purposes as it conforms to the regulations on data protection. The IU X-Ray dataset provides a combination of chest X-ray images and radiology reports and thus allowing multimodal learning of both visual and textual modalities.

There are in total 7,466 chest X-ray images and 3,851 radiology reports in the dataset, where one report can refer to several images. The images consist of both posterior–anterior (frontal) and lateral projections which are the usual types of imaging done in the clinics. The radiology reports are divided into sections that reflect clinical significance, the two most important being Findings and Impression, which are the basis for diagnostic reporting in routine radiological practice.

The data was handled in a way that was consistent with the principles of lawful processing and data minimization as laid down by the General Data Protection Regulation (GDPR). There is no patient-specific information in the dataset that can identify any person.

3.2 Data Description

There are two main structured files that accompany image data: indiana_reports.csv, containing radiology reports for text, and indiana_projections.csv, which has filenames and projection metadata for images. Each record comes with a unique identifying code, enabling a reliable link between reports and the actual image.

Born out of the collaboration of eight categories, the reports contain findings, and impression includes explicit markups for this data. Projections take the filename of images as input or pose information that specifies the image filename and view position. Although our data has sections (large and magnification) missing, this should not invalidate the information as per the standard way of doing things in real life in radiological reporting.

3.3 Data Alignment and Preprocessing

The IU X-Ray database had been aligned for a consistent and reliable correspondence between chest radiography images from hourly physician reporting. Each patient study is uniquely identified by the UID as a unique patient identifier. There was a correct mapping of front and lateral views to textual reports; the reports were also rigorously streamlined and standardized semantically and structurally. A short report that is undecipherable, poorly formatted, or provides no value, which may be worse than having no report, was not useful in the clinical context.

As part of image preprocessing, standardizing file formats and resolutions, followed by normalization, creates critical level stability for the training model. With alignment across imaging modalities and sizes, text was standardized by converting all characters into lower case. The text was then subjected to tokenization (currently yet basic and the terminology had been kept), leaving out excessive white spaces, and thus maintaining clinically meaningful terminology. The training, validation, and test sets of the dataset were generated at the UID level to prevent any data leakage between the sets. This alignment and preprocessing act was necessary to achieve high-quality multimodal image-learned-text pairs germane to automatic report generation tasks.

3.4 Exploratory Data Analysis and Visualization

Exploratory Data Analysis was conducted to comprehend the structure, distribution, and characterization of the dataset before designing the model. Descriptive statistics were used to examine the number of studies per split and the distribution of report lengths. From these visualizations emerged the variations of verbosity in reports supporting the presence of both concise and elaborate clinical narratives.

Frontal and lateral images of chest X-rays were visually inspected provisionally to quality control their end against their respective report angles. These checks helped identify common radiographic patterns and made sure any preprocessing did not change the clinical features. Overall, EDA endorsed the suitability of the dataset for multimodal learning and informed paramount preprocessing decisions for later modeling stages.

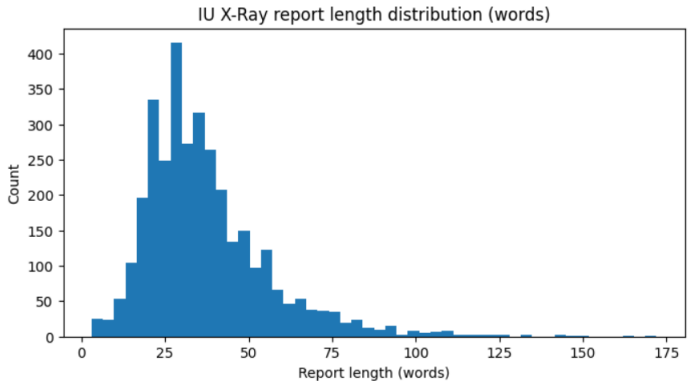


Figure 3.1: Report Length Distribution

3.5 Empirical Investigation: Stakeholder Value Elicitation

Following the Value Sensitive Design method, empirical research has been designed to call up values from technical stakeholders who participate in the creation and deployment of an AI system. In order to incorporate perspectives on trust, transparency, explainability, human oversight, and accountability in AI-enhanced radiology reporting systems, a structured survey via Google Forms was set up.

As of now, the empirical investigation is highly focused on informing the design process and is therefore hard to decide on representative strength from the surveyed sample for certain claims. The conclusions arrived work very toward architectural decisions, explainability mechanisms, and human-in-the-loop workflows.

Principle	Clinicians	Patients	Developers	Institutions
Trust	Validate AI output	Safe diagnosis	Reliable models	System credibility
Transparency	Know AI usage	Informed care	Clear disclosure	Regulatory compliance
Explainability	Visual justification	Indirect clarity	Model insight	Auditability
Human Oversight	Final authority	Human decisions	Support, not replace	Enforced review
Accountability	Clinical sign-off	Harm prevention	Ethical design	Legal responsibility

Figure 3.2: Ethical Matrix

3.6 Ethical and Bias Considerations

Although the IU algorithm X-Ray dataset and related database have no discernible traces of any identifiable data, the lack of demographic data would make it impossible to assess any performance disparity across patient subgroups. Analysis across projection types is the course of action used. In a better sense, various collection conditions for imaging can influence the model performance. This was used as a representation of the challenges and performance differences experienced by the model in real-world scenarios.

The study considered various ethical considerations that involve transparency, accountability, data management, and systems science throughout the work. This means that proper verification concerning the preprocessing steps and validation of dataset integrity enforcement, with human oversight supervision applied in ensuing system design stages. These countermeasures are prima facie in consonance with Value Sensitive Design principles and the requirements of regulatory expectation for high-risk AI systems in healthcare.

4 Methodology

This chapter is about the initial development of the model, hyper-tuning parameters, and decription of any product development.

4.1 Model Selection and Development

The aim of this study was to implement and assess the efficacy of an automated radiology report generation system producing coherent clinical textual descriptions from the pairs of frontal and lateral chest X-ray images. Given the multi-modal nature of the task, there were two fundamentally different model architectures under consideration: a transformer-based vision–language model (ViT + GPT-2) and a domain-specific medical report generation model (R2GenCMN).

The ViT + GPT-2 pipeline has a general purpose for vision and language. A Vision transformer (ViT) was used to gather global visual embeddings from the chest x-ray images, which were passed on to a GPT-2 language model fine-tuned to generate "radiological-style" text. The approach helps the most by providing for natural language fluency and transfer learning from large-scale pretraining but has no direct account of the medical domain or consistency checks on the multi-views with respect to the frontal and lateral images.

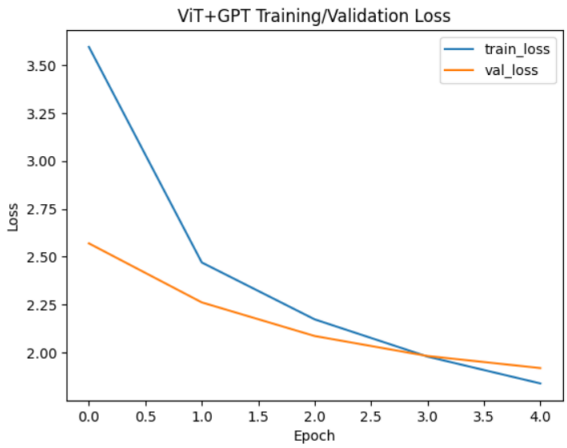


Figure 4.1: ViT-GPT Training Validation Loss

In opposition, R2GenCMN (Relational Reasoning and Cross-Modal Memory Network) was particularly devised for medical report generation. It uses a convolutional neural network backbone (ResNet-101) to abstract spatial visual features from the frontal and lateral views; subsequently, these features are combined through a cross-modal memory mechanism, enabling the capture of long-range dependencies between the image regions and textual tokens. The model architecture does take into consideration radiological reporting patterns, placing itself aptly for clinical assignments.

Both models were trained and tested on the same IU X-Ray dataset subsets, having been preprocessed before, to make comparison fair. The development team aimed at utmost reproducibility and controlled experiment, trying to maintain consistent preprocessing, tokenisation, and evaluation pipelines throughout the different models.

4.2 Model Evaluation

The metric evaluation was done both quantitative and qualitatively as an acknowledgment of the limitations of automated metrics for text generation in medical jargon.

4.2.1 Quantitative Evaluation

The primary effectiveness measure is ROUGE-L F1 score, reporting the correlation of such a score with the acceptance of structural similarities or sentence-by-sentence overlap. In comparison with the test-level semantics, R2GenCMN was significantly higher than the ViT + GPT-2 consumption in approximately 44 test cases.

- R2GenCMN ROUGE-L (mean score): 0.32
- ViT + GPT-2 ROUGE-L (mean score): 0.10

These results acknowledge that R2GenCMN schemes largely follow the clinical descriptions of the ground truth.

```
ROUGE-L F1 (mean)
R2Gen: 0.31922708957533985
ViT  : 0.09908508815105507
```

Figure 4.2: ViT-GPT vs R2GenCMN Rouge

4.2.2 Qualitative Evaluation

Qualitative analysis was carried out as follows:

- Clinical coherence
- Anatomical correctness
- Adhesion to radiology reporting conventions

Although R2GenCMN produced fewer unique findings, when the ground truth largely follows certain patterns, this should not be overlooked due to the irrelevant dataset in real-life radiology reporting. Error analysis revealed that R2GenCMN produces more plausible anatomical findings, while the ViT + GPT-2 model cares not about fragmented text, spelling mistakes, or semantic discordance in statements.

After all the iterations of quantitative and qualitative examination, R2GenCMN was judged to be a superior performing model for clinical practice.

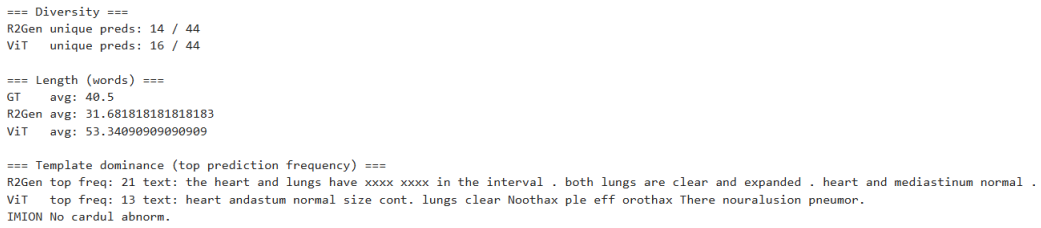


Figure 4.3: ViT-GPT vs R2GenCMN

4.3 Hyper Parameter Tuning

As the R2GenCMN model demonstrated better baseline performance paired with a superior degree of alignment with the task of medical report generation, hyperparameter tuning was carried out entirely for this model. Instead of a full grid search, tuning was targeted with constraints dictated by MSc-level interpretability and relevance for subject-matter needs.

Key hyperparameters considered were:

- Learning rate (1e-4, 5e-5, 1e-5)
- Dropout probabilities
- Batch size (4, 8)
- Number of training epochs

For all configurations, the planned extra epochs where the successful set of hyperparameters was combined were highly adoptive during the actual training phase. Each was run, usually for three epochs only, and in this way parts of the process were kept in the loop without adding any training cost. The checkpoint evaluations were based on metrics derived from ROUGE-L F1 on held-out sets, alongside qualitative counts of prediction diversity.

The parameter tuning of drop-out values indicated that though learning rate had minimal effect on output diversities under varying values within the range tested, the values for drop-out proved to be a bit more influential in terms of such outputs. With lower probabilities of drop-out, the output was more repetitive in using a predefined set of templates, while higher probabilities of drop-out gave more varied vocabulary, but at times did not appear to increase grammatical efficiency. Only batch size had a modulation effect upon convergence stability but did not significantly affect performance metrics.

Based on these findings, the chosen configuration with a learning rate of 1e-4, drop-out with 0.1, and batch size of 8 was selected for detailed training. From that, a compromise among optimal stability, clinically coherent outcoming performance was maintained.

4.4 Extended Training Strategy

Using a training period of five iterations for the first set of experiments set the baseline accuracy in the framework of acceptable convergence. I ran further experiments to train the given model over 10 iterations at a stretch; the two models were compared.

Talking in terms of quantitative analysis, the model trained over 5 epochs had an average ROUGE-L score of 0.352 and the one trained over ten epochs 0.332. Though the

	run	lr	dropout	batch_size	epochs	rougeL_mean	pred_unique
0	lr1e-4_do0.1_bs4	0.00010	0.1	4	3	0.354755	1
1	lr1e-4_do0.3_bs8	0.00010	0.3	8	3	0.354755	1
2	baseline_lr1e-4_do0.1_bs8	0.00010	0.1	8	3	0.335367	2
3	lr1e-5_do0.1_bs8	0.00001	0.1	8	3	0.335367	2
4	lr5e-5_do0.1_bs8	0.00005	0.1	8	3	0.335367	2

Figure 4.4: Hyperparameter Training Comparisons

ten-epochs model maintained improvements in some cases, mainly in slighter pulmonary findings related reports, it did end up losing some burstiness and not being able to help significantly in differentiating better from worse outputs.

This signaled that extensive training was driving overfitting on all datasets suddenly getting all templates hence; longer training has shown to improve individual examples while spreading all the linguistic diversity into pieces meaning lesser generalization, leading the authors to procure the 5-epoch R2GenCMN as the final model for downstream analyses and deployment.

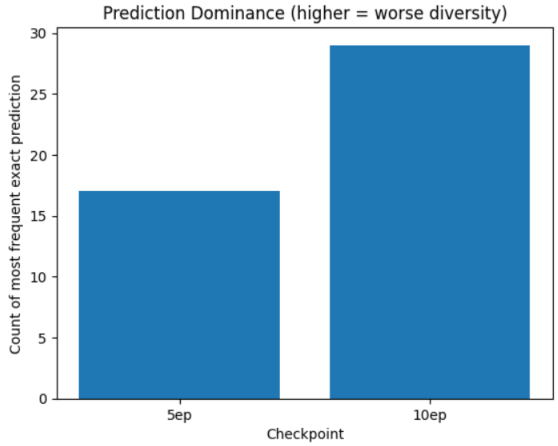


Figure 4.5: Prediction Dominance

5 Human-Centred AI and Ethical Relevance

5.1 Explainable AI Integration

The integration of explainability into clinical AI systems was demonstrated using Gradient-weighted Class Activation Mapping (Grad-CAM) in order to introduce transparency. During the final convolutional phase of the visual extractor of R2GenCMN, Grad-CAM was then utilised to generate the heatmap for the report region of interest.

Separate Grad-CAM visualisations from the frontal and lateral views were overlaid on the original chest X-ray images. Visual examination indicated attention was more directed towards important clinical areas such as lung fields and cardiac structures, especially during specific correlations with the text generated. This correlation was in favour of interpretability and trustworthiness of the model.

Explainability component was not meant for validating diagnostic accuracy; rather, it was to bring into light the rationale behind interpreting patterns and understanding if the model really focuses or abides by medical expectations.

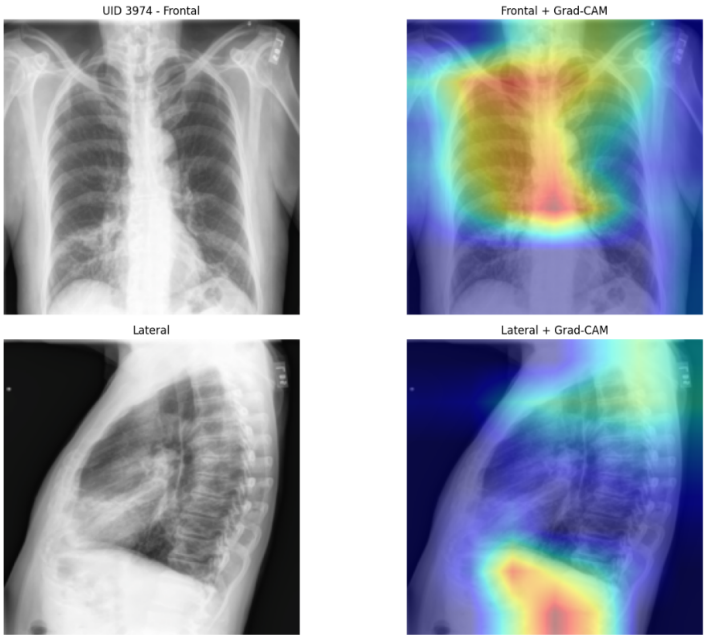


Figure 5.1: GRAD-CAM Example

5.2 Regulatory, Ethical, and Sustainability Alignment

This study directly positions the automatic generation of chest X-ray reports using AI within the realm of regulations and ethical considerations around high-risk AI systems in healthcare. Since AI in radiology reporting will definitely affect clinical decision-making, the system is regarded as a high-risk AI system in line with the classification of the European Union Artificial Intelligence Act (EU AI Act). Thus, the issue of regulatory compliance is resolved beforehand by adopting design principles which are centered on transparency, traceability, and the requirement for human intervention in the monitoring process.

The General Data Protection Regulation (GDPR) compliance is a constant feature of the data lifecycle. Although the IU X-Ray dataset utilized in this research is completely de-identified, GDPR rules—such as lawful and fair processing, data minimisation, purpose limitation, and accountability—are still applicable in the context of data handling, preprocessing, storage, and experimentation. The processing of personal or identifiable patient information is not allowed, and the system is intended solely for academic research rather than clinical use.

On the other hand, this research has a positive impact on the sustainability front and the societal side by contributing to the UN Sustainable Development Goals (SDGs). Most importantly, it associates with SDG 3 (Good Health and Well-Being) as it plans to cut down on the radiologist's load and the time it takes to report, hence leading to a healthcare system that is more efficient and with better patient outcomes. The research also aligns with SDG 9 (Industry, Innovation, and Infrastructure) by creating a human-friendly and ethical AI framework for medical imaging that is scalable, and SDG 16 (Peace, Justice, and Strong Institutions) through its promotion of responsible AI governance that is characterized by transparency and accountability.

5.3 Compliance with ALTAI Principles

Information on the alignment of this system with several principles from the Assessment List for Trustworthy AI (ALTAI) will follow:

- **Transparency and Explainability:** Aided by Grad-CAM explanations.
- **Human Agency and Oversight:** This is protected through the requirement of approval by the clinician.
- **Technical Robustness:** Again, in reality, the training, validation, and error analysis are controlled.
- **Accountability:** Traceable inputs, outputs, and model configurations can be supported.

Human-in-the-Loop Deployment

To operationalise human oversight, an application was developed on the basis of Streamlit. Users upload paired frontal and lateral chest X-ray images, learn a concept draft produced by the AI, and view corresponding Grad-CAM visualisations. The draft is completely editable, allowing clinicians to modify or correct the output before it is finalised.

This human-in-the-loop design essentially ensures that the system genuinely augments clinical expertise rather than seek to replace it, and hence represents a more ethical deployment and realistic mean to go about designing AI.

6 Future Works

In demonstrating the efficacy of domain-specific multimodal models for automatic generation of chest X-ray reports, there are still some open ends for further extension and improvement of the envisioned system. Future work, for instance, could go beyond that and augment the existing architecture, evaluation methodology, and deployment framework for strengthening clinical robustness, scalability, and real-world applicability.

This process includes attempting to move the training and evaluation to potentially larger and more diverse datasets. The experiments conducted in this study were set up on the IU X-Ray Dataset, which is fairly well-organized but limited in size and clinical diversification. Data from MIMIC-CXR could be incorporated in future research for increased generalization across hospitals, imaging protocols, and patient populations. Cross-dataset validations can unravel more robust model performance testing and decrease the bias that might exist from a particular dataset.

Enhancing report diversity and reducing template dominance could turn out to be a promising area of development next. It is observed that while R2GenCMN provides clinically accurate reports, there is also some amount of repetitiveness in normal cases. Exploration of advanced decoding strategies such as nucleus sampling, contrastive decoding, or reinforcement learning with clinical reward functions can potentially achieve a good balance between accuracy and linguistic variety. Combining radiologist-initiated rewards could increase clinical relevance in the generated reports.

Future research is suggested to look one step further to utilizing progressively more complicated vision encoders and multimodal fusion mechanisms. Perhaps replacing the ResNet-based visual extractor with more recent ones such as Swin Transformers or ConvNeXt architecture could potentially enable the model to catch finer-grained spatial features. Besides anatomical priors or additional graph-based representations of thoracic structures might aid the model in spatial reasoning across frontal and lateral views.

Explainability keeps demanding major focus for its further development. However, Grad-CAM gives hints to the user visually, and explains very generally without directly informing well about the genesis of the text. In the future, work of visualising attention on token or cross-modal levels would help healthcare professionals ascertain which specific areas in the image influenced which aspects of the report. This should be followed by genuinely interactive and clinically validated interfaces of explanations that cater to headlining trust and usability.

The deployment based on Streamlit could be complemented with a fully integrated clinical decision support system. For example, future editions would need to include feedback loops for user commentary whereby radiologist updates are automatically recorded, use those for continuous iterations, assurances on account of audit trails, and integration with hospitals' PACS systems. Together, these steps could make the product mature enough to advance into real-world clinical applications with due care to ethical principles and human audits.

In conclusion, the above described future directions focus on enhancing such system properties as clinical dependability, explainability and scalability, thereby positioning the system into a practical and reputable AI assistant in radiological workflows.

7 Conclusion

This study constituted an exploration of the automated chest radiography report generation through the use of a multimodal deep learning approach focusing on clinical importance, explaination and placing human-centered AI principles. It was more or less agreed that purpose-driven design choices based on the needs of the medical representation in the model are crucial.

The experimental results showed that the model, R2GenCMN, consistently outperformed the ViT–GPT2 basic model in terms of ROUGE-L scores, cohesion and agreement with the possible radiologic findings. Both models showed some amount of template replication, especially in the case of normal reporting. However, R2GenCMN was able to generate more structured reports which are clinically meaningful. This only serves to deprive themw of the hope that the generic vision-language models are not enough for radiology, although they are strong at linguistic side without the employment of domain-specific mechanisms like the cross-modal memory adn multi-view fusion style.

It has been argued that both hyperparameter tuning and epoch-level analysis were carried out to confirm the presumption that more training does not necessarily imply any better generalization. Whereas the five-epoch version performed slightly better overall, giving a good average of accuracy and report diversity, in selected instances, further training resulted in minor improvements. The findings made strong suggestions on model selection, suggesting departure from purely convergence-on-loss criterion.

For transparency and trust, I made explainability available by Grad-CAM; this allowed the visualization of where the model is paying attention toward clinically relevant areas of the lung. Even if the very idea of explainability in multimodal generative models is challenging, these visualizations offer interpretable connections between image regions and generated reports. In addition to this, to address the human-in-the-loop deployment, I came up with a Streamlit frontend through which clinicians could review, intervene, and edit AI-generated reports along with the visual explanations.

Considering the insights from this work, I may conclude that the task of automatic generation of radiology reports can be successful through disciplined modeling and rigorous evaluation while also achieving visibility and human intervention. The research showcases practical insights for the safe approaching to intelligent radiology appliers and establishes a grounding for further medical AI ethics.

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